Short-Term Electricity Price Forecast Using Frequency Analysis and Price Spikes Oversampling

Chenxu Zhang, *Student Member, IEEE*, Yong Fu, *Senior Member, IEEE*, and Lin Gong, *Senior Member, IEEE*

***Abstract*—This paper proposes a new short-term electricity price forecast framework using frequency analysis and price spikes oversampling. To perform normal price forecast, firstly, the variational mode decomposition (VMD) method is adopted to decompose price data into multiple frequency band-limited modes where each mode has a center frequency. Then, the extended discrete Fourier transform (EDFT) method is utilized to transform the decomposed price modes from time domain into frequency domain to predict the price. In addition, to facilitate price spike prediction, oversampling methods including the enhanced structure preserving oversampling (ESPO) and the synthetic minority oversampling technique (SMOTE) for regression, are adopted to synthesize price spike cases. Multiple electricity markets are studied and numerous comparison results illustrate that the proposed forecast framework is capable to accurately predict both normal electricity prices and price spikes.**

***Index Terms*—Extended discrete Fourier transform, price spikes, short-term price forecast, variational mode decomposition.**

I. Introduction

SHORT-TERM (day-ahead and hours-ahead) electricity price forecasts are essential to all electricity market participants [1], [2], [3]. For generation companies, accurate short-term price forecasts are used to derive optimal bidding strategies; for load serving entities, accurate short-term price forecasts can assist them to obtain sufficient energy with low costs; also, for trading companies, profitable bidding strategies are achieved if price forecasts are accurate. Due to its importance to all electricity market participants, an accurate short-term electricity price forecast is highly desired in the modern electricity market today [4], [5].

Researchers stated that short-term electricity prices are a type of time-series data that possess both periodic and uncertain characteristics [6]. Daily, weekly, seasonally, and other common periodic patterns were used in price forecasts [7]. Reference [8] applied the weighted nearest neighbors' methodology in the day-ahead price forecast. It is a type of similar day forecast method that assumes that if the prices of the previous day i are close to the prices of the current day d, then the prices of day $i+1$ in the past are similar to the prices of day $d+1$. More seasonal components are considered in forecast model by using the wavelet decomposition or the Hodrick and Prescott filter [9]. To learn non-linear relationships between predictors and targets, which are hard to be captured by the similar day method solely, a hybrid method was presented in [10] that combines the artificial neural network (ANN) with the similar day method. Besides, considering the non-linear relationships, authors extend the work of [9] by using the ANN [11]. Reference [12] also combined the autoregressive integrated moving average (ARIMA) method to model daily and weekly linear patterns with the ANN model to capture nonlinear relationships within electricity prices. Comparing with [12] that assumed the error term had a zero mean and constant variance, reference [1] regarded that electricity prices had a time-varying nature of volatility. Therefore, the generalized autoregressive conditional heteroskedastic (GARCH) method was adopted to model the time-varying volatility of the Spain and California day-ahead electricity prices. By taking the advantages of ARIMA method in modeling stationary linear relationship, GARCH method in modeling non-constant variance, and the adaptive wavelet neural network (AWNN) method in modeling non-linear relationships, reference [13] proposed a hybrid model to forecast the day-ahead locational marginal prices of the PJM market. Comparing with [10], forecast results of the hybrid method in [13] are not only accurate but also stable, which means low forecast error and low error variance. Besides, instead of handling price variance in forecast models, authors adopted the variance stabilizing transformations to preprocess data and improve the ANN forecast accuracy [14]. To ensure predictor relevancy, redundancy, and interaction in the forecast process, reference [15] utilized a feature selection method prior to the neural network forecaster. In [16], the weight and biases of the feed forward neural networks were tuned by using the particle swarm optimization. To enable neural network to explore higher nonlinearities, the stacked several denoising autoencoders (SDA) was used in [17] to form a deep neural network for performing forecast. Reference [18] predicted electricity prices by adopting a generative adversarial network (GAN) based approach that leverages the spatio-temporal correlations in historical data. However, training ANN networks is still a time-consuming procedure. To shorten the training period and maintain forecast accuracy, reference [2] utilized the extreme learning machine (ELM) method to

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substitute ANN. As input weights of the ELM method are only generated once instead of being tuned iteratively, the ELM method can shorten computation time.

Uncertainty is another characteristic of electricity prices and price spikes are one of the most salient situations. Normally, price spikes are rarely occurred and the magnitude of spike price is tens or hundreds of times higher than normal electricity price. Therefore, price spike forecast difficulties reside in two aspects: predict occurrence and magnitude of price spikes. By considering the duration of spikes, reference [19] adopt an ARIMA based autoregressive conditional hazard model to forecast spike occurrence. In [20], authors explicitly focused on forecast incidence of price spikes. The probabilistic neural network (PNN) was used to predict price spike occurrence. Besides, the mutual information (MI) based feature selection method was also applied to identify the most useful input data to reduce data redundancy and improve forecast accuracy. Using intra-hour data and applying a lower spike identification threshold, reference [21] provided a framework to forecast the one-hour ahead spike occurrence. Reference [22] also focused on spike occurrence forecast. And by comparing support vector machine (SVM) with probability classifier, they proved that SVM can provide a more reliable spike occurrence prediction. Besides, the capability of SVM to forecast spike occurrent was also proved in [23] by comparing a broader range of spike forecast methods. On the other hand, there are many researches focusing on spike value prediction. Note that as price spikes do not have a universal definition, different thresholds are used to qualify price spikes [24]. By adopting a consecutive method to roughly classify spikes into coarse categories (e.g., prices between [75, 100], [100, 150] … [500, 2000]), the nearest neighboring hybrid method was applied to predict the final spike values by averaging the k nearest prices in the same category [25]. To provide a comprehensive solution, researchers considered both price spike occurrence and value predictions in their works [26], [26], [28], and [19]. Reference [26] offered a framework that combined the SVM with the Bayesian classifier to improve spike occurrence forecast accuracy. And spike values were estimated by the nearest neighboring method in [25], [26]. Besides, combined with the spike occurrence forecast derived from [20], reference [27] proposed a hybrid neuro-evolutionary system (HNES) to predict the spikes values. The HNES framework adopted the neural networks (NNs) to estimate spike magnitudes and utilized the evolutionary algorithms (EA) to tune NNs parameters. After identifying price spikes, the spike values were estimated by using the neural networks method [28].

To some extent, all methods discussed above can be categorized into three groups: the statistical methods, including the GARCH method [1], similar day method [8], and ARIMA method [19], explicitly express periodicities in formulations; the artificial intelligence methods, which include the ELM [2], ANN [11], [14], [15], [16], SDA [17], GAN [18], PNN [20], and SVM [22], [23], use historical price data to train networks that implicitly estimate periodicities in price data; and, the hybrid methods [10], [12], and [13], leverage the characteristics of both statistical and artificial intelligence methods. However, all these methods estimate periodicities through a trial-and-error process that might overlook some important periodicities and result in forecast accuracy degradation. On the other hand, as price spike occurrence is low and historical data is limited, it is hard for those algorithms to provide reliable price spike predictions. Therefore, to identify periodicities resided in electricity prices and increase price spike forecast accuracy, we propose a new short-term electricity price forecast method to predict both normal electricity prices and price spikes. The main contributions of this paper are summarized as follows:

1) This paper presents a comprehensive framework that is not only to take normal price forecast into consideration, which explicitly adopts the identified frequency information to predict electricity prices, but also able to provide price spike classification and regression results, which utilizes oversampling methods to increase forecast accuracy. The effectiveness of the proposed forecast framework is justified in numerous case studies using multiple real market price data, which includes PJM, MISO, ISO-NE, NYISO, and Spain, and comparing with the forecast results published in the prestigious journals.
2) For normal price forecast, to analyze periodicities within electricity prices and reduce random noise, the variational mode decomposition (VMD) is firstly utilized to decompose the frequency mixed time-series electricity prices into a set of modes where each mode has a central frequency. Then, as the extended discrete Fourier transform (EDFT) could achieve a higher resolution in frequency domain than the discrete Fourier transform and does not assume data patterns outside the observation period, by utilizing the frequency information within each mode, the EDFT is adopted in this paper to perform a better normal price prediction.
3) For price spike forecast, to resolve the prediction difficulty caused by data limitation, this paper exploits synthetic oversampling methods to increase the number of price spike cases and balance the training dataset. The predictors of the spike cases are oversampled by using the enhanced structure preserving oversampling (ESPO) method to ensure that the synthesized cases have similar temporal structures as the real price spike cases. The target values of the spike cases are oversampled by utilizing the synthetic minority oversampling technique (SMOTE) for regression. As a result, the oversampled cases can enhance the training dataset for price spike forecaster to identify spike cases and predict their price values better.

The rest of the paper is organized as follows. Section II presents the proposed short-term electricity price forecast framework. Section III discusses three key techniques adopted in the forecast framework. Numerous case studies are discussed in Section IV to demonstrate the effectiveness of our proposed price forecast framework. Conclusions and future works are summarized in Section V. Note that this paper focuses on the point/deterministic price forecasts and the probabilistic price forecasts will be addressed in our future work.

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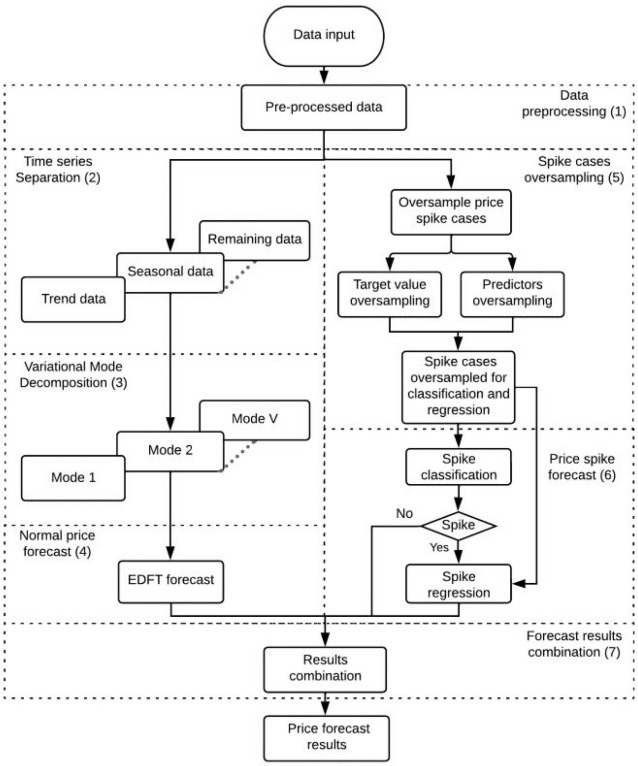

Fig. 1. Framework of the proposed short-term electricity price forecast.

II. Framework of the Proposed Short-Term Electricity Price Forecast

The short-term electricity price forecast framework is shown in Fig. 1. The major steps are described as follows (Note that steps 2-4 are for the normal price forecast and steps 5-6 are for the price spike forecast. They can be implemented in parallel):

Step 1: Data Preprocessing: As the price data directly derived from electricity markets might have deficiencies (e.g., data loss and data duplication), we identify and modify the imperfect data in this step. For example, the missing price data are substituted by the mean values derived from averaging the electricity prices before and after the missing ones; and the duplicated price data are eliminated to ensure every time spot only has one price data.

Step 2: Time-series Separation: Electricity prices can be viewed as a set of time-series data that consists of trend, seasonal, and remainder components [23]. To improve forecast accuracy and facilitate data analysis, we perform a time-series decomposition to separate the trend and seasonal components from the input price data. The trend is calculated by using the moving average method; the seasonal component is captured by averaging the detrended time-series data for a certain seasonality period; and the remainder are derived from the input price data after extracting trend and seasonal components.

Step 3: Variational Mode Decomposition: The price data components (e.g., trend, seasonal, and remainder) discovered from Step 2 are still mixed with frequencies. To further improve prediction accuracy, the frequencies hidden in the decomposed time-series data should be identified. In this step, the VMD method is adopted to transform each price data component into multiple price modes that facilitate the following normal price forecast module.

Step 4: Normal Price Forecast: As the EDFT method is able to discover the frequencies within electricity prices and applies the identified frequencies to price forecast, we adopt the EDFT to forecast the normal electricity price in this step. Since the time-series price decomposition and the VMD decomposition are additive models, the outputs of this step are the summation of the EDFT forecasts for all modes.

Step 5: Spike Cases Oversampling: As the number of electricity price spikes is much less than normal prices and the imbalanced dataset will influence the accuracy of price spike forecasts, we synthesize price spike cases in this step. We first define price spikes for each hour as electricity prices at different hours have different patterns. Then, we oversample price spike cases where each case consists of a target variable and a set of predictors.

Step 6: Price Spike Forecast: As we need to identify price spikes and then predict their values, the aims of this step are related to price spike classification and regression. In order to improve classification correctness and regression accuracy, both the classification and regression methods are trained by using the oversampled dataset derived from step 5.

Step 7: Forecast Results Combination: This step combines the normal price and price spike prediction results. If a forecasted price is classified as a spike, we get the electricity price predictions from the price spike forecasts in step 6. Otherwise, the price values from the normal price forecasts in step 4 are applied. The outputs of the short-term electricity price forecast framework are the combined forecast results.

III. Proposed Key Forecasting Methodologies

In this section, three key methodologies in the proposed price forecast framework, including VMD, EDFT, and spike oversampling methods, are discussed in detail.

A. Variational Mode Decomposition Module

As illustrated in Fig. 1, the VMD [29] method is adopted to decompose the frequency mixed price component data $f(t)$ into several modes where each mode is a set of band-limited data with a center frequency. The input of the VMD module is the trend, seasonal, or remaining data and the outputs are decomposed modes of each component.

Firstly, we transfer the price data into its analytical form as this form has two features that benefit frequency analysis. The first feature is that the data in the analytical form has a positive unilateral spectrum in frequency domain. Therefore, when demodulating price data within frequency domain by shifting frequency spectrum, we only need to consider the shifting effect on a one-sided spectrum. Besides, data in analytical form are easy to restore back to time-series form by retrieving the real part from the analytical form.

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By using (1), the data $u_v(t)$ of price mode v is able to be transformed into its analytical form $u_{A,v}(t)$,

$$u_{A,v}\ (t) = u_v\ (t) + jHu_v\ (t) = \left(\delta\left(t\right) + \frac{j}{\pi t}\right) * u_v\left(t\right), \tag{1}$$

where H is the Hilbert transform operator (the impulse response is $h\ (t) = 1/\pi t$ in time domain and $h\ (\omega) = j\omega/|\omega|$ in frequency domain); $\delta(t)$ is the Dirac delta function; and $*$ is a convolution operator.

After analytical transformation, to derive a band limited price mode $u_v(t)$ in frequency domain, we can recenter the spectrum of $u_v(t)$ and apply a low pass filter. The amount of shift for every mode v is determined by each mode's center frequency ω_v. The formulation of recenter process is shown in (2).

$$u_{A,v}\left(t\right)\ e^{-j\omega_v t} = \left[\left(\delta\left(t\right) + \frac{j}{\pi t}\right) * u_v\left(t\right)\right]\ e^{-j\omega_v t}, \tag{2}$$

Note that after the Fourier transform, the $e^{-j\omega_v t}$ term in time domain is converted to a Dirac delta $\delta(\omega + \omega_v\ )$ in frequency domain. Therefore, in (2), multiplying the analytical form of mode $u_{A,v}(t)$ by $e^{-j\omega_v t}$ in time domain equals to convoluting $u_{A,v}(\omega)$ with $\delta(\omega + \omega_v\ )$ in frequency domain. The effect of this convolution is to shift the spectrum $u_{A,v}(\omega)$ to the left by ω_v amount.

The bandwidth of the price mode $u_v(t)$ can be determined by optimizing the Gaussian smoothness of the demodulated spectrum in frequency domain which equals to optimize the squared L^2 norm of the demodulated data's gradient in time domain. Thus, the objective is formed in (3). The constraint (4) is used to ensure the price component data $f(t)$ can be recovered from the decomposed price modes $u_v(t)$.

$$\min\left\{\sum_v \left\|\partial_t\left[\left(\delta\left(t\right) + \frac{j}{\pi t}\right) * u_v\left(t\right)\right] e^{-j\omega_v t}\right\|_2^2\right\} \tag{3}$$

$$s.t.\ \sum_v u_v\left(t\right) = f\left(t\right). \tag{4}$$

In order to relieve the constraint (4), the augmented Lagrangian method is used to convert the above constrained optimization problem to an unconstrained form (5) by moving the constraint (4) into the objective (3)

$$\begin{aligned}\mathcal{L}\ (u_v, \omega_v, \lambda) \ =\ & \alpha\sum_v \left\|\partial_t\left[\left(\delta\left(t\right) + \frac{j}{\pi t}\right) * u_v\left(t\right)\right] e^{-j\omega_v t}\right\|_2^2 \\ & + \left\|f\left(t\right) - \sum_v u_v\left(t\right)\right\|_2^2 + \lambda\left[f\left(t\right) - \sum_v u_v\left(t\right)\right],\end{aligned} \tag{5}$$

where α is the balancing parameter of the data-fidelity constraint and λ is Lagrangian multiplier in time domain.

Since it is hard to solve the above optimization problem that contains exponential term multiplication, convolution, and derivative operation in time domain, we transfer (5) into frequency domain by using the Fourier transform, where the exponential term multiplication becomes the frequency spectrum shift operation; the convolution becomes multiplication; the derivative operation becomes multiplying the objective function with $j\omega$; and the sgn means the sign function.

$$\begin{aligned}\min\ \alpha\sum_v j\omega\left\|\left[\left(1 + sgn\left(\omega + \omega_v\right)\right) u_v\left(\omega + \omega_v\right)\right]\right\|_2^2 \\ + \left\|f\left(\omega\right) - \sum_v u_v\left(\omega\right)\ + \frac{\lambda\left(\omega\right)}{2}\right\|_2^2.\end{aligned} \tag{6}$$

As each mode in (6) can be viewed as an independent frequency spectrum $u_v(\omega)$ with a center frequency ω_v in frequency domain, by using the alternate direction method of multipliers (ADMM) method, the spectrum of mode $u_v(\omega)$ is calculated by iteratively optimizing (7)

$$\begin{aligned}u_v^{r+1}\left(\omega\right) = \underset{u_v}{\operatorname{argmin}}\{\ & \alpha\left\|j\omega\left[\left(1 + sgn\left(\omega + \omega_v^r\right)\right) u_v^r\left(\omega + \omega_v^r\right)\right]\right\|_2^2 \\ & + \left\|f\left(\omega\right) - \sum_{l\ =\ 1}^{v-1} u_l^{r+1}\left(\omega\right)\right. \\ & \left. - \sum_{l\ =\ v}^{V} u_l^r\left(\omega\right) + \frac{\lambda^r\left(\omega\right)}{2}\right\|_2^2\Bigg\},\end{aligned} \tag{7}$$

where r represents iteration number; l is mode index; and V is the total number of modes. As the price mode data are a set of real numbers, by applying the Hermitian symmetry, the (7) can be simplified to (8) where ω is used to substitute $\omega - \omega_v$ in the first term,

$$\begin{aligned}u_v^{r+1}\left(\omega\right)\ =\ \underset{u_v}{\operatorname{argmin}}\Bigg\{ & \int_0^\infty 4\alpha(\omega - \omega_v^r)^2\left|u_v^r\left(\omega\right)\right|^2 \\ & + 2\left|f\left(\omega\right) - \sum_{l\ =\ 1}^{v-1} u_l^{r+1}\left(\omega\right)\right. \\ & \left. - \sum_{l\ =\ v}^{V} u_l^r\left(\omega\right) + \frac{\lambda^r\left(\omega\right)}{2}\right|^2 d\omega\Bigg\}.\end{aligned} \tag{8}$$

The optimal solution of (8) is calculated by letting its derivative equal to zero. Thus, the $u_v^{r+1}(\omega)$ can be expressed as,

$$u_v^{r+1}\ \left(\omega\right) = \frac{f(\omega) - \sum_{l\ =\ 1}^{v-1} u_l^{r+1}(\omega) - \sum_{l\ =\ v+1}^{V} u_l^r(\omega) + \frac{\lambda^r(\omega)}{2}}{1 + 2\alpha(\omega - \omega_v^r)^2}. \tag{9}$$

By applying (9), the price mode $u_v(\omega)$ is demodulated from the input by using a low-pass narrow-band filter around $\omega\ = \omega_v$ which is able to remove high frequency noise from the data.

As the center frequency ω_v does not appear in the second term of (6), we ignore this term from the center frequency calculation. So, the center frequency ω_v of $u_v(\omega)$ is the optimal solution of (10).

$$\omega_v^{r+1} = \underset{\omega_v}{\operatorname{argmin}}\left\{j\omega\left\|\left[\left(1 + sgn\left(\omega + \omega_v^r\right)\right) u_v^{r+1}\left(\omega + \omega_v^r\right)\right]\right\|_2^2\right\}. \tag{10}$$

Similar to (8), the (10) can be simplified into (11)

$$\omega_v^{r+1}\ =\ \underset{\omega_v}{\operatorname{argmin}}\left\{\int_0^\infty \left(\omega - \omega_v^r\right)^2\left|u_v^{r+1}\left(\omega\right)\right|^2 d\omega\right\}. \tag{11}$$

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The solution of (11) is

$$\omega_v^{r+1} = \frac{\int_0^\infty \omega \left| u_v^{r+1}(\omega) \right|^2 d\omega}{\int_0^\infty \left| u_v^{r+1}(\omega) \right|^2 d\omega}. \tag{12}$$

Note that Lagrangian multipliers $\lambda(\omega)$ in frequency domain is updated by using (13), where τ is the penalty parameter.

$$\lambda^{r+1}(\omega) = \lambda^r(\omega) + \tau \left(f(\omega) - \sum_v u_v^{r+1}(\omega) \right). \tag{13}$$

The above iterative calculations (9), (12) and (13) will run until the convergence criterion (14) is met.

$$\frac{\sum_v \left\| u_v^{r+1}(\omega) - u_v^r(\omega) \right\|_2^2}{u_v^r(\omega)_2^2} \leq \varepsilon. \tag{14}$$

As the output of the VMD module is a set of modes in time domain, by using real part of the inverse Fourier transform in (15), we can transfer $u_v(\omega)$ from frequency domain back to time domain, $u_v(t)$.

$$u_v(t) = Real\left(ift\left(u_v(\omega)\right)\right), \tag{15}$$

where, the $Real$ is an operator that only takes the real part of the inverse Fourier transform ift.

B. Extended Discrete Fourier Transform Module

In the proposed price forecast framework, the EDFT [30] method is selected to analyze and predict normal electricity prices rather than the DFT method. The first deficiency of the DFT method is that the frequency resolution is limited by the length of input. If frequency resolution is not fine enough, the spectrum leakage situation may happen and the true periodicity within price data may not be identified. Besides, the DFT method assumes that data outside the observation period are the same as the data within the period. On the contrary, the EDFT method ensure high frequency resolution around periodicities within the price data without imposing limitations on data outside the observation period. According to Fig. 1, the input of the EDFT module is the time series price mode derived from the VMD module and its outputs are the forecasts for each mode.

As the continuous frequency spectrum derived from the Fourier transform provides the finest frequency resolution, we regard the continuous frequency spectrum as the target and use a new basis $\alpha(\omega,\ t_k)$ to convert data from time domain into frequency domain. To simplify expressions in this section, the symbol $u(t)$ is utilized to substitute the price mode data $u_v(t)$ from (15). Since the price mode data $u(t)$ are sampled uniformly, the basis $\alpha(\omega,\ t_k)$ can be written as $\alpha(\omega, kT)$. So, the EDFT can be written as (16)

$$F_\alpha(\omega) = \sum_{k=0}^{K-1} u(kT)\ \alpha(\omega,\ kT), \tag{16}$$

where $u(kT)$ is the mode derived from the VMD module uniformly and $k = 0, 1, 2 \ldots, K-1$; K is the number of sampled points within the observation period; T is the sampling interval; and ω is the radian frequency.

In order to ensure that the EDFT (16) and the Fourier transform (17) are as close as possible, we formulate the objective (18) where Δ is the squares error expression between (17) and (16).

$$F(\omega) = \int_{-\infty}^{+\infty} u(t)\, e^{-j\omega t}\, dt \tag{17}$$

$$\Delta = \left| F(\omega) - F_\alpha(\omega) \right|^2. \tag{18}$$

However, the Fourier transform (17) cannot be used directly on the input $u(t)$ as there are no infinite long prices period. To circumvent this obstacle, the input prices with ω_0 frequency can be expressed in the form of (19), where $-\infty < t < \infty$

$$u(\omega_0, t) = S(\omega_0)\, e^{j\omega_0 t}. \tag{19}$$

By substituting (19) into (17), the Fourier transform $F(\omega)$ at the frequency ω_0 can be expressed by the Dirac delta function (20)

$$F(\omega_0) = \int_{-\infty}^{+\infty} u(\omega_0, t)\, e^{-j\omega t}\, dt = 2\pi S(\omega_0)\, \delta(\omega - \omega_0), \tag{20}$$

where ω_0 is the cyclic frequency within the range $-\Omega \leq \omega_0 \leq \Omega$; Ω is the upper frequency of the prices $f(\omega_0, t)$; and $S(\omega_0)$ is the complex amplitude of $f(\omega_0, t)$ in frequency domain. By using the $F(\omega)$ in (20) and the $F_\alpha(\omega)$ in (16) where the $u(kT)$ is substituted by using (19), the difference between the Fourier transform and the EDFT is shown in (21).

$$D = 2\pi S(\omega_0)\, \delta(\omega - \omega_0) - \sum_{k=0}^{K-1} S(\omega_0)\, e^{j\omega_0 kT} \alpha(\omega,\ kT). \tag{21}$$

So, the objective function (18) can be rewritten as (22).

$$\Delta = \int_{-\Omega}^{\Omega} |D|^2 d\omega_0. \tag{22}$$

To specifically calculate the EDFT basis $\alpha(\omega,\ kT)$ so as to minimize square error Δ, we can solve (22) by letting the derivative of this function with respect to $\alpha(\omega,\ lT)$ equals to zero, $\partial\Delta/\partial\alpha(\omega,\ lT) = 0$, $l = 0, 1, \ldots K-1$. The solution is shown in (23).

$$\int_{-\Omega}^{\Omega} \left[2\pi S(\omega_0)\, \delta(\omega - \omega_0) - \sum_{k=0}^{K-1} S(\omega_0)\, e^{j\omega_0 kT} \alpha(\omega, kT) \right] \times S(\omega_0)^* e^{-j\omega_0 lT} d\omega_0 = 0. \tag{23}$$

As Dirac delta functions are able to select frequencies, the (23) can be simplified into (24) where the first term of (23) is at the right side of (24).

$$\sum_{k=0}^{K-1} \left(\frac{1}{2\pi} \int_{-\Omega}^{\Omega} |S(\omega_0)|^2 e^{j\omega_0 (k-l)T} d\omega_0 \right) \alpha(\omega,\ kT) = |S(\omega)|^2 e^{-j\omega lT}. \tag{24}$$

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The solutions of (24) that is the EDFT basis $\alpha(\omega,\ kT)$ can be expressed (25)

$$\boldsymbol{A}_{\omega} = |S(\omega)|^2\ \boldsymbol{R}^{-1}\boldsymbol{E}_{\omega}, \tag{25}$$

where $\boldsymbol{E}_{\omega}$: $e^{-j\omega lT}$, and the components of $\boldsymbol{R}(K \times K)$, are expressed as (26).

$$r_{l,k} = \frac{1}{2\pi}\int_{-\Omega}^{\Omega} |S(\omega_0)|^2 e^{j\omega_0(k-l)T} d\omega_0 \tag{26}$$

Then, we can utilize (27) to transform the input price mode in time domain into frequency domain by using the new derived basis.

$$F_{\alpha}\ (\omega) = \ \boldsymbol{u}\ \boldsymbol{A}_{\omega} = |S(\omega)|^2\ \boldsymbol{u}\boldsymbol{R}^{-1}\boldsymbol{E}_{\omega},\ -\Omega \le \omega \le \Omega \tag{27}$$

To facilitate the above calculation, continuous function integration can be substituted by summation. For instance, the (26) can be estimated by (28) if the number of sampled points N within the integral interval is large enough

$$r_{l,k} \approx \frac{\Omega}{\pi N}\sum_{n=0}^{N-1} |S(\omega_n)|^2 e^{j\omega_n(k-l)T}, \tag{28}$$

where $-\Omega \le \omega_{\mathrm{n}} \le \Omega\ n = 0, 1, \ldots N-1$. To derive (28) from (26), we can regard the integration as to calculate the area under a curve. The inner part of integration is the height and $d\omega_0$ is the width. In order to substitute integration with summation, we have $2\Omega/N$ points between frequency interval $[-\Omega,\ \Omega]$.

By viewing (27), the last obstacle that hinders the EDFT calculation is that the true amplitude spectrum $S(\omega)$ of the input price mode is unknown. As suggested by [31], the amplitude spectrum can be obtained by using an iterative process (29)-(32). The (29) is the matrix form of (28). The (30) is the estimation of (27). And the amplitude spectrum $S(\omega)$ is estimated by (31). The weight vector $\boldsymbol{W}$ is calculated by (32)

$$\boldsymbol{R}^i = \frac{1}{\boldsymbol{N}}\boldsymbol{E}\boldsymbol{W}^i\boldsymbol{E}^H \tag{29}$$

$$\boldsymbol{F}^i = \boldsymbol{u}\ \boldsymbol{A}^i = \boldsymbol{u}\left(\boldsymbol{R}^i\right)^{-1}\boldsymbol{E}\boldsymbol{W}^i \tag{30}$$

$$\boldsymbol{S}^i = \frac{\boldsymbol{u}\left(\boldsymbol{R}^i\right)^{-1}\boldsymbol{E}}{diag\left(\boldsymbol{E}^H\boldsymbol{R}^i\boldsymbol{E}\right)} \tag{31}$$

$$\boldsymbol{W}^{i+1} = diag\left(\left|\boldsymbol{S}^i\right|^2\right) \tag{32}$$

where superscript i is iteration number and $\boldsymbol{W}^1 = \boldsymbol{I}$ for the first iteration. The superscript H stands for Hermitian transpose. $\boldsymbol{E}(K \times N)$ has elements $e^{-j\omega_n kT}$that is a vector form of $\boldsymbol{E}_{\omega}$ where continuous frequency ω is estimated by using ω_{n}. The above iteration stops if the result of $\boldsymbol{S}^i$ at the current iteration is close to the last iteration.

Now, the output of the EDFT module $\boldsymbol{u}_e$ can be extrapolated. As the number of frequency component N is larger than the number of input data K, by using the inverse discrete Fourier transform (IDFT) (33), we can obtain a time sequence $\boldsymbol{u}_e$ that contains a number of $\mathrm{N}-\mathrm{K}$ extrapolated data.

$$\boldsymbol{u}_e = \frac{1}{N}\boldsymbol{F}\boldsymbol{E}_N^H \tag{33}$$

where $\boldsymbol{E}_N(N \times N)$ is a matrix that has element $e^{-j2\pi f_n mT}$, $m = 0, 1 \ldots,\ N-1$ and $\omega_n = 2\pi f_n$.

C. Price Spike Oversampling Module

As the low occurrence rate of price spike affects classifiers to recognize spike patterns and undermines regression models to forecast spike values, the price spike oversampling module is adopted in the proposed forecast framework to increase the number of price spike cases. The ESPO method [32] is used to oversample predictors while preserving temporal relationship among data and the SMOTE for regression method [33], [34] is adopted to oversample the targets.

To create the predictors of spike cases using the ESPO method, we firstly calculate the spike case predictors covariant matrix $\boldsymbol{W}$ by using (34).

$$\boldsymbol{W} = \frac{1}{N_{case}}\sum_{i=1}^{N_{case}}\left(\boldsymbol{P}_i - \bar{\boldsymbol{P}}\right)\left(\boldsymbol{P}_i - \bar{\boldsymbol{P}}\right)^T \tag{34}$$

where $\bar{\boldsymbol{P}} = 1/N_{case}\ \sum_{i=1}^{N_{case}}\boldsymbol{P}_i$ represents the mean of spike case predictors and $\boldsymbol{W}$ is a positive semi-definite covariant matrix. Then, we utilize the eigen decomposition (35) on $\boldsymbol{W}$.

$$\boldsymbol{D} = \boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V} \tag{35}$$

where $\boldsymbol{D}$ is the diagonal eigenvalue matrix with descending order; the $\boldsymbol{V}$ is the eigenvector matrix of $\boldsymbol{W}$; and the superscript T means transposition. Eigenvectors of a covariance matrix can be viewed as data varying directions and eigenvalues are the magnitude of variance on the direction. In order to create new spike cases that have a similar temporal structure as the real spike cases, in the eigenvector directions derived from (35), the eigenvalues of synthetic cases should be comparable with the eigenvalues of real cases.

Although the eigenvalue matrix $\boldsymbol{D}$ can be measured from data, to facilitate data synthesis, it is convenient to scale the eigenvector matrix $\boldsymbol{V}$ and convert $\boldsymbol{D}$ to an identity matrix $\boldsymbol{I}$. Therefore, we scale the eigenvector matrix $\boldsymbol{V}$ using (36) to divide each eigenvector v_k by $\sqrt{d_k}$, where d_k is the eigenvalue of the eigenvector v_k.

$$\boldsymbol{F} = \boldsymbol{V}\boldsymbol{D}^{-1/2} \tag{36}$$

Then, the (35) can be rewritten as (37).

$$\begin{aligned}\boldsymbol{I} &= \boldsymbol{D}^{-\frac{1}{2}}\ \boldsymbol{V}^T\boldsymbol{W}\boldsymbol{V}\ \boldsymbol{D}^{-\frac{1}{2}} = \boldsymbol{F}^T\ \boldsymbol{W}\boldsymbol{F} \\ &= \frac{1}{N_{case}}\sum_{i=1}^{N_{case}}\boldsymbol{F}^T\left(\boldsymbol{P}_i - \bar{\boldsymbol{P}}\right)\left(\boldsymbol{P}_i - \bar{\boldsymbol{P}}\right)^T\boldsymbol{F} \\ &= \frac{1}{N_{case}}\sum_{i=1}^{N_{case}}\left(\boldsymbol{F}^T\boldsymbol{P}_i - \boldsymbol{F}^T\bar{\boldsymbol{P}}\right)\left(\boldsymbol{F}^T\boldsymbol{P}_i - \boldsymbol{F}^T\bar{\boldsymbol{P}}\right)^T\end{aligned} \tag{37}$$

If we define

$$\boldsymbol{Z} = \boldsymbol{F}^T\left(\tilde{\boldsymbol{P}} - \bar{\boldsymbol{P}}\right) \tag{38}$$

$\tilde{\boldsymbol{P}}$ are the synthesized predictors. In this paper, we generate $\boldsymbol{Z}$ from multivariate Gaussian distribution $N(0, \boldsymbol{I})$ as a large number of random and independent distributions that obey the

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Gaussian distribution [35]. Therefore, if we generate a sufficient number of rare cases, after eigen decomposition and scaling, the eigenvalue matrix of the generated cases covariance matrix also has an identify matrix structure, which ensures the synthesized predictors have the same temporal structure as the original rare cases. After getting $\boldsymbol{Z}$, the synthesized predictors $\tilde{\boldsymbol{P}}$ can be derived from (39).

$$\tilde{\boldsymbol{P}} = \boldsymbol{V}\boldsymbol{D}^{\frac{1}{2}}\boldsymbol{Z} + \bar{\boldsymbol{P}} \tag{39}$$

After synthesizing predictors, the next step is to synthesize the target value of spike case. The SMOTE for regression method [34], as a weighted average method where the larger distance has a smaller weight, is used in this step to calculate the values of the synthesized target. First, select a spike case with the synthesized predictor derived by using (39) and find its R nearest spike cases in the real spike case dataset. Then, calculate the distance between the predictor and its R nearest spike cases. Finally, adopt the (40) to calculate the value of its target G_s

$$G_s = \frac{\sum_{r=1}^{R} G_r L_{rs}}{\sum_{r=1}^{R} L_{rs}} \tag{40}$$

where L_{rs} is the Euclidean distance between predictor values of real case r and synthetic case s, and G_r is the target values of the R nearest spike cases in the real spike case dataset.

IV. Case Studies

In this section, the following electricity price data sources are used to show the performance of the proposed forecast framework for both normal price and price spike predictions:

- Pennsylvania-New Jersey-Maryland Interconnection (PJM) [36]
- Midcontinent Independent System Operator (MISO) [37]
- ISO New England (ISO-NE) [38]
- New York Independent System Operator (NYISO) [39]
- Day ahead electricity market of mainland Spain [40]

For the PJM market, case studies utilize the electricity price data from the PJM-RTO node. The hourly day-ahead electricity prices of the MISO market are collected from the Indiana hub. For the ISO-NE market, the hourly locational marginal prices (LMP) are from the ISO-NE hub. The day-ahead market locational based marginal pricing (LBMP) of the NYISO market is selected from the N.Y.C. zone. And, for the Spain electricity market, the day-ahead hourly prices of mainland Spain are used.

Besides, to leverage the latest price information for forecasts, we adopt a rolling window scheme to perform forecasts. The normal price forecaster is calibrated using the price data in a three-month time window prior to the forecast date. When prices of the next 24 hours are predicted, the calibration window rolls forward 24 hours. This rolling procedure ends until the price prediction of the last day is completed. And, for price spike forecast, we also adopt the rolling window scheme. However, as price spikes rarely occurred, the calibration window are extended to include the price data of up to two previous years.

TABLE I
Comparisons of Daily AMAPES

Date in 2006	Prop.	Ref. [13]	Improv.	Ref. [10]	Improv.
Jan. 20	2.61%	3.71%	29.6%	6.93%	62.3%
Feb. 10	3.12%	2.85%	-9.47%	7.96%	58.9%
Mar. 5	2.53%	5.48%	53.8%	7.88%	67.9%
Apr. 7	3.15%	4.17%	24.5%	9.02%	65.1%
May 13	3.27%	4.06%	20.5%	6.91%	52.7%

TABLE II
Comparison of Daily Error Variances

Date in 2006	Prop.	Ref. [13]	Improv.	Ref. [10]	Improv.
Jan 20	0.00070	0.0010	30.0%	0.0034	79.4%
Feb 10	0.00078	0.0050	84.4%	0.0050	84.4%
Mar 5	0.00043	0.0033	87.0%	0.0061	93.0%
Apr 7	0.00051	0.0013	60.8%	0.0038	86.6%
May 13	0.00043	0.0015	71.3%	0.0049	91.2%

A. Normal Electricity Forecast Results

In this subsection, we use the price data from the PJM to test the performance of normal price forecast. The chosen dates include January 20, February 10, March 5, April 7, and May 13 in the year 2006 [10], [13]. To measure the price forecast accuracy, the average mean absolute percentage error (AMAPE) metric is used to compared forecast performance of the proposed method and the results published in [10] and [13]. The comparison results are listed in Table I. The first column indicates the forecast date. The AMAPE values of our proposed forecast method, [13], and [10] are listed in the column 2, 3, and 5, respectively. A lower AMAPE value means more accurate forecast results. To facilitate comparison, columns 4 and 6 list the forecast accuracy improvements of our predictions compared with the forecasts of [13] and [10], respectively. A positive value means that the predictions using our forecast method are more accurate than the forecasts reported in the references. Also, we can tell that the AMAPE values of our proposed method are around 3%. Compared with [10], our method has an overall better performance as the AMAPE values of [10] are basically above 6%. Besides, the forecast accuracy of our method is also better than [13] except for the predictions on February 10. However, the AMAPE value of our forecasts is very close to the [13], which are 3.12% and 2.85%. The largest AMAPE improvement of our proposed method compared with [13] and [10] are 53.8% and 67.9% on March 5. Table II shows the variance of forecast errors, σ^2. A smaller variance means a more stable forecast performance. The variance of the proposed forecast method is between 0.00043 on March 5 and 0.00078 on February 10. Compared with [13] and [10], the proposed method errors variance is overall lower. The largest variance improvements of our method compared with [13] and [10] are 87% and 93% on March 5, respectively.

To better understand the functionality of key modules in normal price forecast, predictions on March 5, 2006 are selected as an example. According to the proposed forecast framework, the first step is to read data and perform data preprocessing.

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TABLE III
ELECTRICITY PRICES STATISTICS FROM JANUARY 2, 2005 TO JULY 31, 2006

Hour	Threshold ($/MWh)	# of price spikes	# of normal prices	Spike/normal ratio
1	58.60	29	549	5.28%
2	53.15	21	555	3.78%
3	48.40	25	549	4.55%
4	47.92	26	550	4.73%
5	48.85	29	547	5.30%
6	60.72	28	548	5.11%
7	103.69	20	556	3.60%
8	105.43	21	555	3.78%
9	99.48	20	556	3.60%
10	99.30	24	552	4.35%
11	100.03	35	541	6.47%
12	109.61	22	554	3.97%
13	122.72	19	557	3.41%
14	129.57	20	556	3.60%
15	135.30	23	553	4.16%
16	143.57	23	553	4.16%
17	153.82	20	556	3.60%
18	164.43	10	566	1.77%
19	145.07	14	562	2.49%
20	132.54	23	553	4.16%
21	122.48	24	552	4.35%
22	104.16	28	548	5.11%
23	75.98	31	545	5.69%
24	65.85	25	551	4.54%

TABLE IV
CONFUSION MATRIX DERIVED FROM THE SVM TRAINED BY THE ORIGINAL DATASET

Items	True Spike	True Normal
Spike Prediction	38	47
Normal Prediction	32	3,531

TABLE V
CONFUSION MATRIX DERIVED FROM THE SVM TRAINED BY THE OVERSAMPLED DATASET

Items	True Spike	True Normal
Spike Prediction	65	42
Normal Prediction	5	3,536

TABLE VI
CLASSIFICATION PERFORMANCE COMPARISON

	Original data	Oversampled data	Improvement
Precision	44.71%	60.75%	35.88%
Recall	54.29%	92.85%	71.03%
F2-score	0.52	0.84	61.31%

The imported historical prices are from December 4, 2005 to March 4, 2006 and no missing or duplicated data are found. The time series decomposition is the second step. As the variance of electricity prices doesn't increase with time, we adopt an additive model to separate the trend, seasonal, and remainder data, which is shown in Fig. 2. Since the price data components derived from step 2 are still mixed with frequencies, the third step of our proposed forecast procedure is to adopt the VMD decomposition to transform each price data component into multiple modes. We separate the trend data into four modes, the seasonal data into one mode, and the remainder data into four modes. As an example, the modes of the remainder data in the training period are shown in Fig. 3. The fourth step is to use the EDFT to perform forecast for each mode. Fig. 4 shows the forecast results of the remaining data. Obviously, the forecasts for each mode, except for mode 2, perfectly trace their real values. Although the forecast accuracy of mode 2 is not good enough, comparing the electricity prices that are normally above 40 $/MWh, it doesn't have a significant impact on the overall forecast accuracy as it only account for a tiny portion of the total price. The final step is to compose the forecast results. As we adopt the additive model, forecast results of the trend, seasonal, and remainder data/modes are added together to obtain the final normal price predictions, which is shown in Fig. 5. Compared with the AMAPEs reported in [13] and [10] that are 5.48% and 7.88%, respectively, the AMAPE of our price forecasts is 2.53%, which means we have more accurate prediction results.

TABLE VII
PRICE SPIKE FORECAST RESULTS ON AUGUST 2, 2006

Hour	Actual Price ($/MWh)	Forecast Price ($/MWh)	Relative Error
1	99.85	100.21	0.36%
2	80.00	82.48	3.10%
3	63.55	68.47	7.73%
10	143.33	157.69	10.02%
11	169.11	176.99	4.66%
12	208.12	208.86	0.35%
13	235.87	231.26	1.95%
14	271.30	264.38	2.55%
15	301.82	293.76	2.67%
16	321.45	313.06	2.61%
17	328.05	319.03	2.75%
18	298.60	287.65	3.67%
19	265.06	251.31	5.18%
20	217.38	202.60	6.80%
21	208.92	196.84	5.78%
22	183.45	178.46	2.71%
23	136.79	142.87	4.45%
24	94.12	112.60	19.64%

TABLE VIII
AMAPE OF ELECTRICITY MARKETS

Date in 2020	MISO	ISO-NE	NYISO	Spain
Mar. 1-7	5.06%	8.46%	4.40%	6.62%
Jun. 1-7	5.20%	5.96%	4.92%	4.35%
Sep. 1-7	3.77%	6.48%	4.02%	3.72%
Dec. 1-7	3.56%	7.07%	3.79%	4.54%
Average	4.40%	6.99%	4.28%	4.81%

TABLE IX
VARIANCE OF ERRORS OF ELECTRICITY MARKETS

Date in 2020	MISO	ISO-NE	NYISO	Spain
Mar. 1-7	0.0014	0.0042	0.0011	0.0034
Jun. 1-7	0.0019	0.0028	0.0027	0.0014
Sep. 1-7	0.0010	0.0035	0.0009	0.0009
Dec. 1-7	0.0008	0.0026	0.0008	0.0014
Average	0.0013	0.0033	0.0014	0.0018

B. Electricity Price Spike Forecast Results

In this subsection, to evaluate the performance of price spikes classification and regression, the PJM price data from January 2,

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TABLE X
RMAE OF ELECTRICITY MARKETS

Date in 2020	MISO			ISO-NE			NYISO			Spain		
	LEAR	DNN	Prop.	LEAR	DNN	Prop.	LEAR	DNN	Prop.	LEAR	DNN	Prop.
Mar. 1-7	0.52	0.62	0.29	0.85	0.62	0.45	0.82	1.04	0.49	0.88	0.92	0.55
Jun. 1-7	0.73	0.73	0.28	1.08	0.73	0.47	0.64	0.75	0.29	0.74	0.94	0.34
Sep. 1-7	0.89	0.75	0.28	0.69	0.75	0.38	0.64	0.78	0.27	0.59	0.59	0.49
Dec. 1-7	0.95	0.91	0.37	0.83	0.91	0.42	0.70	1.00	0.33	0.77	0.75	0.53
Average	0.77	0.75	0.31	0.86	0.75	0.43	0.70	0.89	0.34	0.74	0.80	0.48

TABLE XI
P-VALUE OF THE GW TEST OF ELECTRICITY MARKETS

Comparison	MISO	ISO-NE	NYISO	Spain
LEAR : Prop.	3.89e-07	2.41e-06	9.57e-06	7.65e-04
DNN : Prop.	2.75e-06	5.27e-06	4.80e-07	9.36e-05

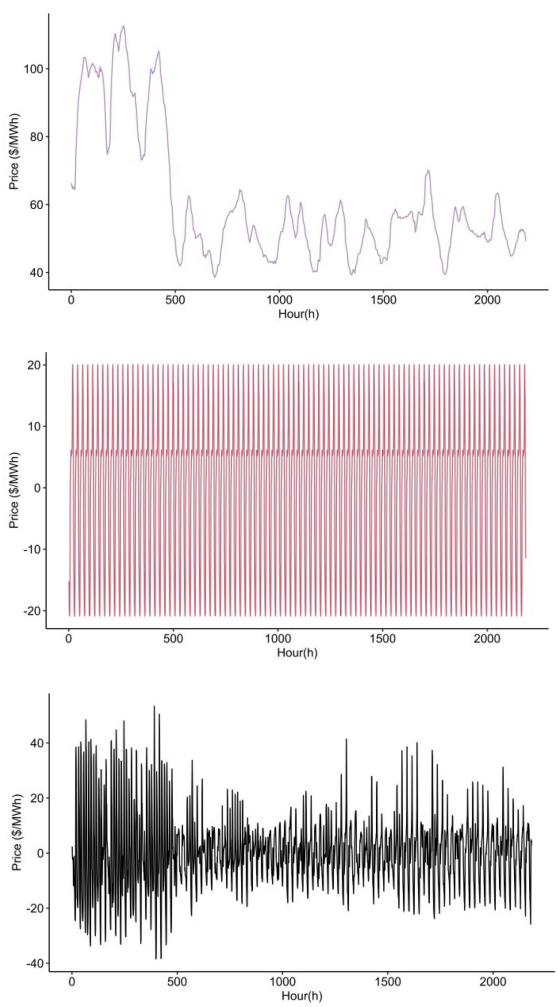

Fig. 2. Time series decomposition of the imported PJM electricity price for the period between December 4, 2005 and March 4, 2006. Trend (top), seasonal (middle), and the remaining data (bottom).

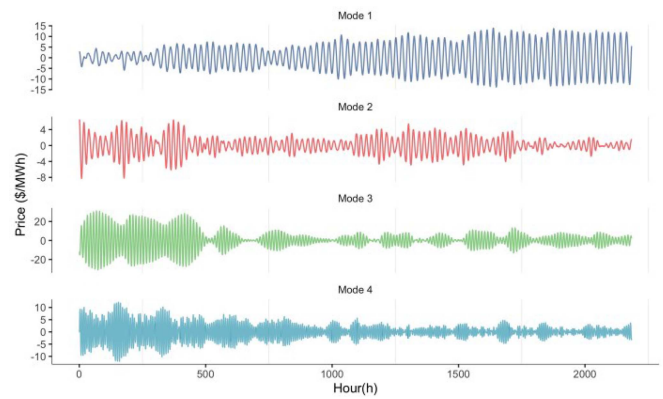

Fig. 3. VMD decomposed remaining data of the PJM electricity price in the training period for the period between December 4, 2005 and March 4, 2006.

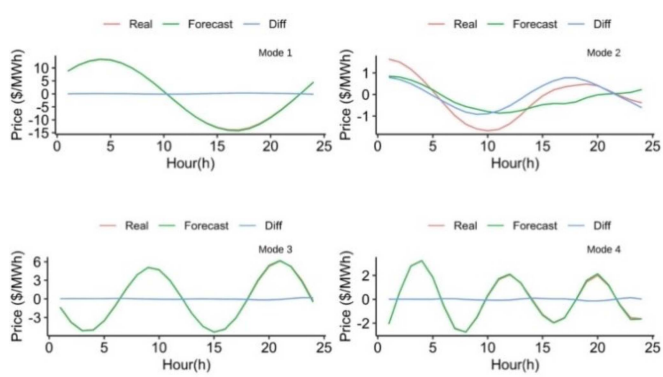

Fig. 4. EDFT forecast for the remaining data of the PJM electricity price on March 5, 2006.

2005 to July 31, 2006 are studied. As there is no universal price spike definition [24], we need to define what is price spike first. In [21], electricity prices more than threshold $\mu + 2\sigma$ are defined as price spikes, where μ is the mean of the historical prices and σ is the standard deviation. On the other hand, authors [27] regard prices above 150 $/MWh or 200 $/MWh as price spikes. However, the $\mu + 2\sigma$ definition will be heavily influenced by large prices as the mean value is sensitive to extremes, and the fixed thresholds are too rigid and may not hold for certain markets or circumstances. Therefore, we adopt the interquartile range (IQR) method to identify price spikes [41] as quantiles are able to tolerant extreme values and more flexible than the fixed threshold method.

The electricity price statistics that are derived from the IQR method are shown in Table III. Data in the shaded area are related to the on-peak hours and otherwise are related to the off-peak hours. The first column is the time of a day. The second column shows the upper-level threshold for each hour. The prices larger than or equal to the thresholds are regarded as price spikes. The

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TABLE XII
AMAPE/MAPE OF ELECTRICITY MARKETS

Market	Node	Testing Date	Prop. AMAPE	Ref. [15] AMAPE	Improv.	Prop. MAPE	Ref. [16] MAPE	Improv.
Spanish electricity markets	Spain	Feb. 18-24, 2002	3.14%	3.36%	6.55%	3.18%	4.04%	21.29%
		May 20-26, 2002	3.06%	3.84%	20.31%	3.35%	3.44%	2.62%
		Aug. 19-25, 2002	5.53%	5.96%	7.21%	5.42%	5.33%	-1.69%
		Nov. 18- 24, 2002	5.62%	5.74%	2.09%	5.78%	6.52%	11.35%
MISO	Indiana hub	**Testing Date**	**Prop. MAPE**	**Ref. [17] MAPE**	**Improv.**	**Prop. MAPE**	**Ref. [18] MAPE**	**Improv.**
		Jan. 1-31, 2014	7.99%	29.78%	73.17%	7.99%	23.76%	66.37%
		Jul. 1-31, 2014	5.42%	8.97%	39.58%	5.42%	8.13%	33.33%

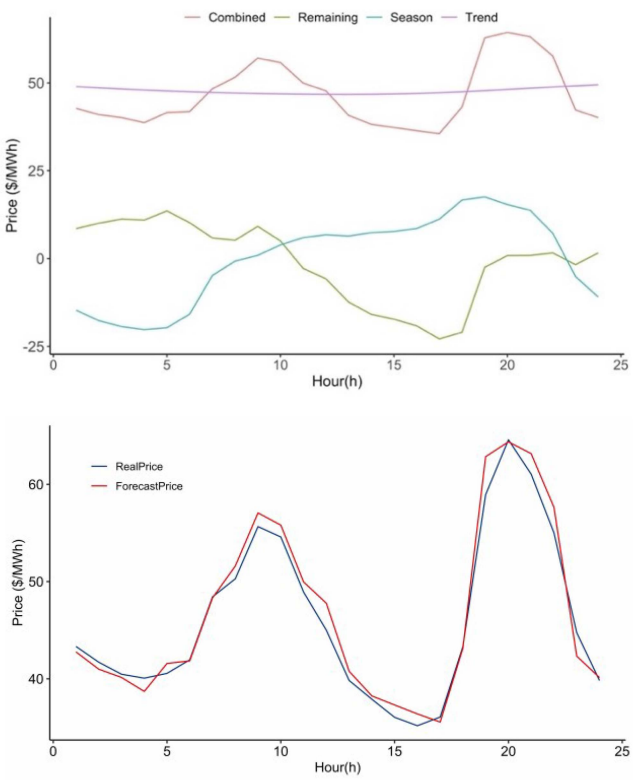

Fig. 5. The trend, seasonal, remaining, and combined day-ahead electricity price forecast of the PJM day-ahead market (top), the real and predicted electricity prices (bottom), on March 5, 2006.

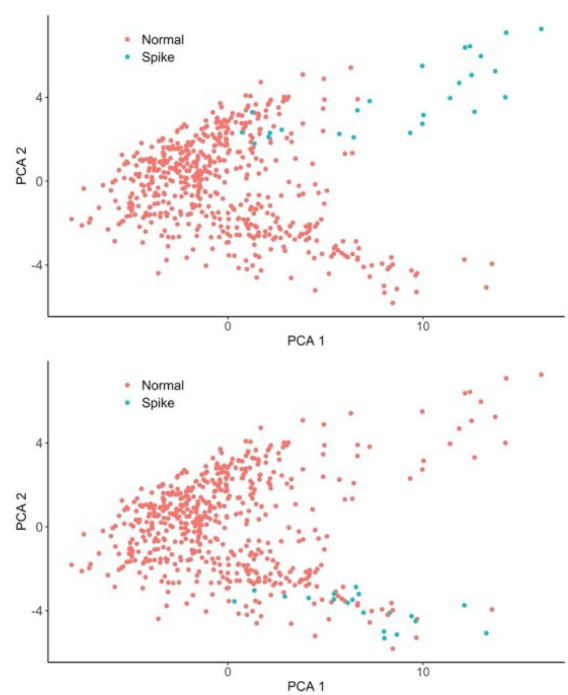

Fig. 6. Scatter plot of the original dataset at hour 4 (top) and hour 17 (bottom).

third and fourth columns list the number of price spikes and normal electricity prices for each hour, respectively. And the last column provides the ratio of price spikes to normal prices. According to Table III, the number of price spikes is far less than the number of normal prices, which the spike/normal ratios are normally below 10% for each hour. Therefore, to derive a balanced dataset to facilitate price spike forecast, we adopt the oversampling method to increase the price spike cases. The predictors and target value of price spike cases are synthesized by using the ESPO and SMOTE, respectively.

Besides, as electricity prices at different hours have different patterns, the oversampling methods are applied to the dataset of each hour. As examples, we oversample the dataset of two hours. One of the selected hours is hour 4 that has the lowest price spike threshold according to Table III and the other is hour 17. To visualize the effectiveness, the results before and after oversampling are plotted in Figs. 6 and 7, respectively, where the x-axis and y-axis are the principal components calculated by using the principal component analysis (PCA) method. As shown in Fig. 6, at hour 4, although the price spike cases in the original dataset are separated from the normal price cases, the total amount of price spike cases is much less than the normal price cases. At hour 17, it becomes hard to differentiate the price spike cases from the normal price cases as they are mixed together. After oversampling, from Fig. 7, the number of price spikes at hour 4 increases and the boundary between the normal price cases and the price spike cases is clearer than the original dataset. For hour 17, after oversampling, although some of the normal price cases are still mixed with the price spike cases, compared with the original dataset, it is much easier to differentiate those two types of cases in the oversampled dataset.

To evaluate the effect of oversampling on price spike classification, we compared the forecast results derived from the classifiers that are trained by using the original/oversampled dataset. Since the performance was well proved in many publications [22], [23] and [32], we adopt the support vector machines (SVM)

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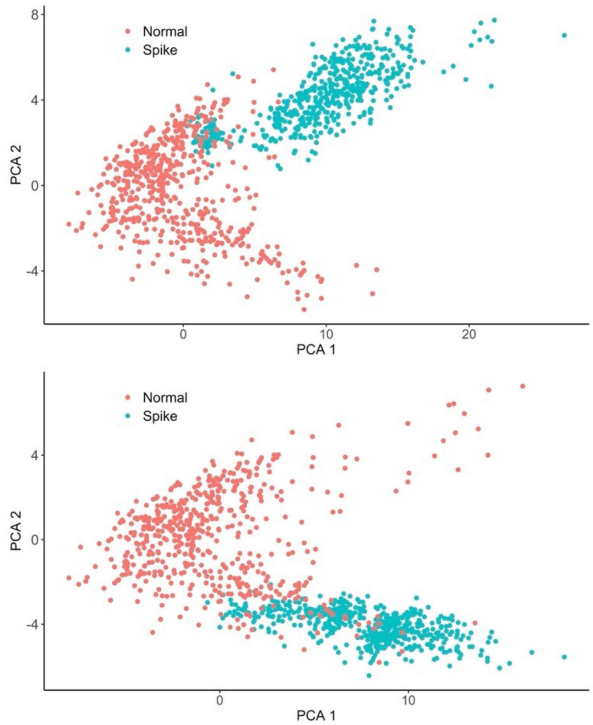

Fig. 7. Scatter plot of the oversampled dataset at hour 4 (top) and hour 17 (bottom).

as the classifier. The training period is from January 2, 2005 to July 31, 2006 and the test period is from August 1, 2006 to December 30, 2006.

Table IV shows the confusion matrix derived from the SVM trained by the original dataset. The precision is $38/(38 + 47) = 44.71\%$ and the recall is $38/(38+32) = 54.29\%$. The F2-score is 0.52. On the other hand, Table V shows the confusion matrix derived from the SVM trained by the oversampled dataset. The precision is $65/(65+42) = 60.75\%$ and the recall is $65/(65+5) = 92.85\%$. The F-2 score is 0.84. The detailed comparisons are summarized in Table VI. Since the recall metric measures the ability of a classifier to identify price spikes, we care more about the recall improvement. From Table VI, we can tell that the recall improvement is 71.03% by using the proposed method. Besides, as we adopt the F2-score to assess the overall performance of a classifier, around 60% improvement means our proposed method could significantly improve price spike classification.

After identifying price spikes, the regression model trained by the oversampled dataset is used to predict their values. As the highest day-ahead electricity price in 2006 occurred on August 2, to show our forecast performance, we use prices within that date as examples. The numeric predictions are listed in Table VII. Note that as price data within hour 4-9 on August 2 are not identified as price spikes, they are not shown in Table VII and should be obtained from the normal price forecast module. At hour 17 which has the highest price in 2006, our forecasted price is 319.03 \$/MWh and the relative forecast error is 2.75%, whereas the price spike prediction in [13] is 289.83 \$/MWh and the relative forecast error is 11.65%. Therefore, by adopting the proposed oversampling method, we can reduce the relative error 76.39%.

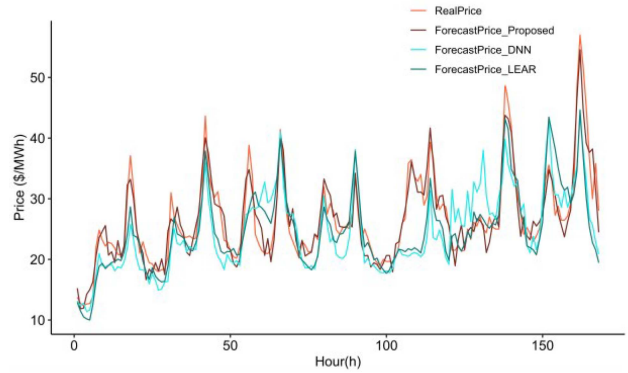

Fig. 8. Real and predicted electricity prices of ISO-NE in December.

C. Price Forecast Results in Multiple Electricity Markets

To further demonstrate the effectiveness of our proposed price forecast framework, multiple electricity markets including the MISO, ISO-NE, NYISO, and the electricity market of Spain are studied. For each market, the case studies will predict the day-ahead electricity prices in the first week of each season in 2020: Mar 1-7 (Spring), Jun. 1-7 (Summer), Sep. 1-7 (Fall), and Dec. 1-7 (Winter). To show the performance of our price forecasts, the AMAPE and variance of forecast errors are listed in Tables VIII and IX, respectively. From Table VIII, we can tell that for each market, our proposed forecast performances for the selected weeks are consistent and acceptable. And, the variances of forecast errors listed in Table IX also show that the proposed price forecast performance is stable for all testing markets under different situations.

To provide further assessments, we also use the relative mean absolute error (rMAE) as a metric to measure the accuracy of price forecasts as the paper [42] comments that the scaled error metric can facilitate comparison between datasets. Besides, the Giacomini-White (GW) test is adopted to determine whether the predictions derived from the proposed method are statistically accurate than the forecasts derived from the benchmark models. The open-access benchmark models [42] we adopt are the Lasso Estimated Auto Regression (LEAR) and the Deep Neural Network (DNN) where the LEAR is a linear statical model and the DNN is able to model the nonlinear relationships between the predictors and the target variables. As shown in Table X, the rMAE values of our proposed approach are constantly lower than the benchmark methods for the selected markets/seasons. Besides, to determine whether the difference of forecast accuracy is statistically significant, the p-values of the GW test are listed in Table XI. As the p-values close to 0 means there are significant differences between forecast results, combined with the accuracy results shown in Table X, we can tell that our price forecasts are statistically better than the predictions derived from the benchmark models. Besides, to provide a visual comparison, predictions of the ISO-NE in December and NYISO in September are plotted in Figs. 8 and 9, respectively. As shown

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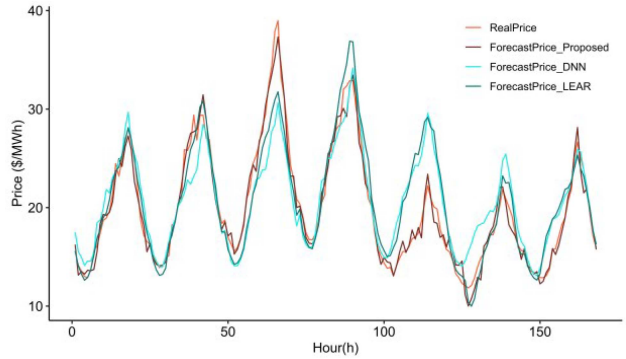

Fig. 9. Real and predicted electricity prices of NYISO in September.

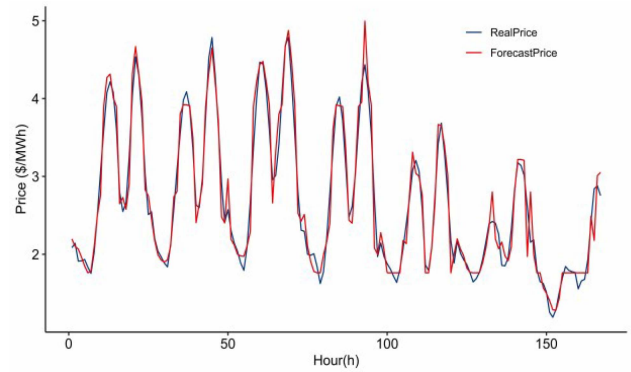

Fig. 10. Real and predicted electricity prices of Spanish electricity market in November.

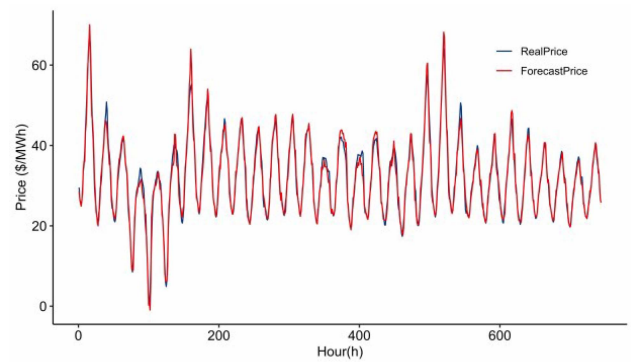

Fig. 11. Real and predicted electricity prices of MISO in July.

in the plots, the curve of proposed predictions is closer to the real price curve than the curves obtained by benchmark models.

D. Forecast Comparisons With References [15], [16], [17], [18]

In this subsection, the forecast results derived from our proposed method are compared with the prediction results provided in more recent publications [15], [16], [17], [18]. As discussed in the introduction section, the methods in [15], [16], [17], [18] implicitly exploit the periodicities within price data by using the neural network based forecast frameworks, which might overlook some important periodicities and result in prediction accuracy degradation. The effectiveness of our proposed method, which explicitly adopts the identified frequency information to predict electricity prices, is further shown in this comparison study.

To provide a clear comparison, the used datasets, the proposed forecast results, the published price predictions, and the performance improvements of the proposed method over the benchmark methods are listed in Table XII. The improvements demonstrate that our proposed method could provide more accurate forecast results than the referred methods. Particularly, for the MISO market, as price spikes occurred in January, the accuracies of referred results in [17] and [18] are a little bit low. Compared with references [17] and [18], the forecast accuracy improvements of our proposed method are 73.17% and 66.37%, respectively. To visualize the forecast results, the real and predicted electricity prices of the Spanish electricity markets in November and MISO in July are plotted in Figs. 10, 11, respectively. It can be observed that the forecasts derived from the proposed method are able to effectively capture different real electricity prices patterns.

V. Conclusion

An accurate short-term electricity price forecast is highly demanded for electricity market participants to hedge against risks and arbitrage from electricity markets. In this paper, we presented a comprehensive framework to accurately forecast short-term electricity prices including both normal prices and price spikes. Our framework integrates the VMD method with the EDFT method to reduce random noise and accurately predict normal prices. Also, to increase price spike forecast accuracy, both ESPO and SMOTE algorithms are used to oversample price spike cases. Numerous case studies for different electricity markets are utilized to assess the overall performance of our proposed framework. By using the accuracy and statistical metrics to evaluate, our proposed framework is capable to predict short-term electricity prices with acceptable accuracy and reliable performance. The developed accurate price forecast tool can be used to provide generation companies optimal bidding strategies, to enable customers to obtain energy with low costs, as well as to facilitate trading companies to create arbitrages.

References

[1] R. C. Garcia, J. Contreras, M. van Akkeren, and J. B. C. Garcia, "A GARCH forecasting model to predict day-ahead electricity prices," *IEEE Trans. Power Syst.*, vol. 20, no. 2, pp. 867–874, May 2005.

[2] X. Chen, Z. Y. Dong, K. Meng, Y. Xu, K. P. Wong, and H. W. Ngan, "Electricity price forecasting with extreme learning machine and bootstrapping," *IEEE Trans. Power Syst.*, vol. 27, no. 4, pp. 2055–2062, Nov. 2012.

[3] F. J. Nogales, J. Contreras, A. J. Conejo, and R. Espinola, "Forecasting next-day electricity prices by time series models," *IEEE Trans. Power Syst.*, vol. 17, no. 2, pp. 342–348, May 2002.

[4] T. Hong, P. Pinson, Y. Wang, R. Weron, D. Yang, and H. Zareipour, "Energy forecasting: A review and outlook," *IEEE Open Access J. Power Energy*, vol. 7, pp. 376–388, 2020.

[5] R. Weron, "Electricity price forecasting: A review of the state-of-the-art with a look into the future," *Int. J. Forecasting*, vol. 30, no. 4, pp. 1030–1081, 2014.

[6] S. Chai, Z. Xu, and Y. Jia, "Conditional density forecast of electricity price based on ensemble ELM and logistic EMOS," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 3031–3043, May 2019.

Authorized licensed use limited to: UNIVERSITY OF BIRMINGHAM. Downloaded on November 19,2025 at 21:54:43 UTC from IEEE Xplore. Restrictions apply.

[7] J. Nowotarski, J. Tomczyk, and R. Weron, "Robust estimation and forecasting of the long-term seasonal component of electricity spot prices," *Energy Econ.*, vol. 39, pp. 13–27, 2013.
[8] A. T. Lora, J. M. R. Santos, A. G. Exposito, J. L. M. Ramos, and J. C. R. Santos, "Electricity market price forecasting based on weighted nearest neighbors techniques," *IEEE Trans. Power Syst.*, vol. 22, no. 3, pp. 1294–1301, Aug. 2007.
[9] J. Nowotarski and R. Weron, "On the importance of the long-term seasonal component in day-ahead electricity price forecasting," *Energy Econ.*, vol. 57, pp. 228–235, 2016.
[10] P. Mandal, T. Senjyu, N. Urasaki, T. Funabashi, and A. K. Srivastava, "A novel approach to forecast electricity price for PJM using neural network and similar days method," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 2058–2065, Nov. 2007.
[11] G. Marcjasz, B. Uniejewski, and R. Weron, "On the importance of the long-term seasonal component in day-ahead electricity price forecasting with NARX neural networks," *Int. J. Forecasting*, vol. 35, no. 4, pp. 1520–1532, 2019.
[12] G. P. Zhang, "Time series forecasting using a hybrid ARIMA and neural network model," *Neurocomputing*, vol. 50, pp. 159–175, 2003.
[13] L. Wu and M. Shahidehpour, "A hybrid model for day-ahead price forecasting," *IEEE Trans. Power Syst.*, vol. 25, no. 3, pp. 1519–1530, Aug. 2010.
[14] B. Uniejewski, R. Weron, and F. Ziel, "Variance stabilizing transformations for electricity spot price forecasting," *IEEE Trans. Power Syst.*, vol. 33, no. 2, pp. 2219–2229, Mar. 2018.
[15] O. Abedinia, N. Amjady, and H. Zareipour, "A new feature selection technique for load and price forecast of electrical power systems," *IEEE Trans. Power Syst.*, vol. 32, no. 1, pp. 62–74, Jan. 2017.
[16] Anamika, R. Peesapati, and N. Kumar, "Electricity price forecasting and classification through wavelet–dynamic weighted PSO–FFNN approach," *IEEE Syst. J.*, vol. 12, no. 4, pp. 3075–3084, Dec. 2018.
[17] L. Wang, Z. Zhang, and J. Chen, "Short-term electricity price forecasting with stacked denoising autoencoders," *IEEE Trans. Power Syst.*, vol. 32, no. 4, pp. 2673–2681, Jul. 2017.
[18] Z. Zhang and M. Wu, "Predicting real-time locational marginal prices: A GAN-based approach," *IEEE Trans. Power Syst.*, vol. 37, no. 2, pp. 1286–1296, Mar. 2022.
[19] T. Christensen, A. Hurn, and K. Lindsay, "Forecasting spikes in electricity prices," *Int. J. Forecasting*, vol. 28, no. 2, pp. 400–411, 2012.
[20] N. Amjady and F. Keynia, "Electricity market price spike analysis by a hybrid data model and feature selection technique," *Electric Power Syst. Res.*, vol. 80, no. 3, pp. 318–327, 2010.
[21] H. Chitsaz, P. Zamani-Dehkordi, H. Zareipour, and P. P. Parikh, "Electricity price forecasting for operational scheduling of behind-the-meter storage systems," *IEEE Trans. Smart Grid*, vol. 9, no. 6, pp. 6612–6622, Nov. 2018.
[22] J. H. Zhao, Z. Y. Dong, X. Li, and K. P. Wong, "A framework for electricity price spike analysis with advanced data mining methods," *IEEE Trans. Power Syst.*, vol. 22, no. 1, pp. 376–385, Feb. 2007.
[23] J. H. Zhao, Z. Y. Dong, and X. Li, "Electricity market price spike forecasting and decision making," *IET Gener., Transmiss. Distrib.*, vol. 1, no. 4, pp. 647–654, Jul. 2007.
[24] J. Janczura, S. Trück, R. Weron, and R. C. Wolff, "Identifying spikes and seasonal components in electricity spot price data: A guide to robust modeling," *Energy Econ.*, vol. 38, pp. 96–110, 2013.
[25] X. Lu, Z. Y. Dong, and X. Li, "Electricity market price spike forecast with data mining techniques," *Electr. Power Syst. Res.*, vol. 73, no. 1, pp. 19–29, 2005.
[26] J. H. Zhao, Z. Y. Dong, X. Li, and K. P. Wong, "A general method for electricity market price spike analysis," in *Proc. IEEE Power Eng. Soc. Gen. Meeting*, 2005, vol. 1, pp. 286–293.
[27] N. Amjady and F. Keynia, "A new prediction strategy for price spike forecasting of day-ahead electricity markets," *Appl. Soft Comput.*, vol. 11, no. 6, pp. 4246–4256, 2011.
[28] H. S. Sandhu, L. Fang, and L. Guan, "Forecasting day-ahead price spikes for the Ontario electricity market," *Electric Power Syst. Res.*, vol. 141, pp. 450–459, 2016.
[29] K. Dragomiretskiy and D. Zosso, "Variational mode decomposition," *IEEE Trans. Signal Process.*, vol. 62, no. 3, pp. 531–544, Feb. 2014.
[30] V. Liepins, "Extended Fourier analysis of signals," Mar. 2019. [Online]. Available: https://doi.org/10.48550/arXiv.1303.2033
[31] V. Liepins, "A method of spectrum evaluation applicable to analysis of periodically and non-regularly digitized signals," *Autom. Control Comput. Sci.*, vol. 27, no. 6, pp. 46–52, Dec. 1993.
[32] H. Cao, X. Li, D. Y. Woon, and S. Ng, "Integrated oversampling for imbalanced time series classification," *IEEE Trans. Knowl. Data Eng.*, vol. 25, no. 12, pp. 2809–2822, Dec. 2013.
[33] R. Ribeiro, "Utility-based regression," Ph.D. dissertation, Dept. C.S., Univ. of Porto, Porto, Portugal, 2011.
[34] L. Torgo, R. P. Ribeiro, B. Pfahringer, and P. Branco, "Smote for regression," in *Proc. Prog. Artif. Intell.*, 2013, pp. 378–389.
[35] X. Jiang, "Linear subspace learning-based dimensionality reduction," *IEEE Signal Process. Mag.*, vol. 28, no. 2, pp. 16–26, Mar. 2011.
[36] PJM Web Site, Accessed: 2021. [Online]. Available: http://www.pjm.com
[37] MISO Web Site, Accessed: 2021. [Online]. Available: https://www.misoenergy.org/
[38] ISO-NE Web Site, Accessed: 2021. [Online]. Available: https://www.iso-ne.com/
[39] New York, NY, USAISO Web Site, Accessed: 2021. [Online]. Available: https://www.nyiso.com/
[40] Market Operator of the Electricity Market of Mainland Spain, OMEL, Accessed: 2021. [Online]. Available: https://www.omie.es/en/sobre-nosotros
[41] R. H. Lock, *Statistics: Unlocking the Power of Data*. Hoboken, NJ, USA: Wiley, 2013.
[42] J. Lago, G. Marcjasz, B. De Schutter, and R. Weron, "Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark," *Appl. Energy*, vol. 293, 2021, Art. no. 116983.

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